



CAT 2025

MBA FASTRACK

Lecture - 7

ALGEBRA

MODULUS

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#Q. $f(x) = f(x - 1) + x$, and $f(1) = 6$, find $f(10)$.

- A** 20
- B** 30
- C** 50
- D** 60 ✓✓

$$\left\{ \begin{array}{l} f(x) - f(x-1) = x \\ f(10) - f(9) = 10 \\ f(9) - f(8) = 9 \\ \vdots \\ f(2) - f(1) = 2 \end{array} \right.$$

$$f(10) - f(1) = 54$$

$$f(10) = 54 + 6 = 60 \checkmark$$

#Q. $f(x) - f(x-1) = f(x)f(x-1)$, and $f(1) = 0.1$, find $f(10)$.

A 1 ✓

$$\frac{\cancel{f(x)}}{\cancel{f(x)} f(x-1)} - \frac{\cancel{f(x-1)}}{f(x) \cancel{f(x-1)}} = 1 \quad \text{HINTS}$$

B 0.1

$$\frac{1}{f(x-1)} - \frac{1}{f(x)} = 1$$

$$\frac{1}{f(1)} - \frac{1}{f(10)} = 9$$

C 2

$$\frac{1}{\cancel{f(9)}} - \frac{1}{f(10)} = 1$$

D 0.2

$$\frac{1}{f(1)} - \frac{1}{\cancel{f(2)}} = 1$$

TOPICS *to be covered*



Topic

Modulus : Meaning and Basics

Topic

$|x - a| + |x - b|$ form

Topic

Modulus - Miscellaneous



Topics : Modulus – Meaning and Basics

$$|-8| = 8$$

$$|8| = 8$$

$$|-x| = x$$

What is Modulus ??



$$|8| = (8) = 8$$

QUANTITY INSIDE MODULUS ≥ 0

$$(8) = 8$$

$$| \quad | \rightarrow (\quad)$$

$$|2| = (2) = 2$$

$$|-3| = -(-3) = 3$$

$$|-8| = 8$$

QUANTITY INSIDE MODULUS < 0

$$| \quad | \rightarrow -(\quad)$$

$$\underbrace{|-8|}_{\text{neg}} \rightarrow -(-8) \rightarrow 8$$

What is Modulus ??



$|a - b| \rightarrow$ Dist b/n 'a' & 'b'

Dist b/n x & 3 is 8

$$|x - 3| = 8$$

ve

$$x - 3 = 8$$

$$\boxed{x = 11} \checkmark$$

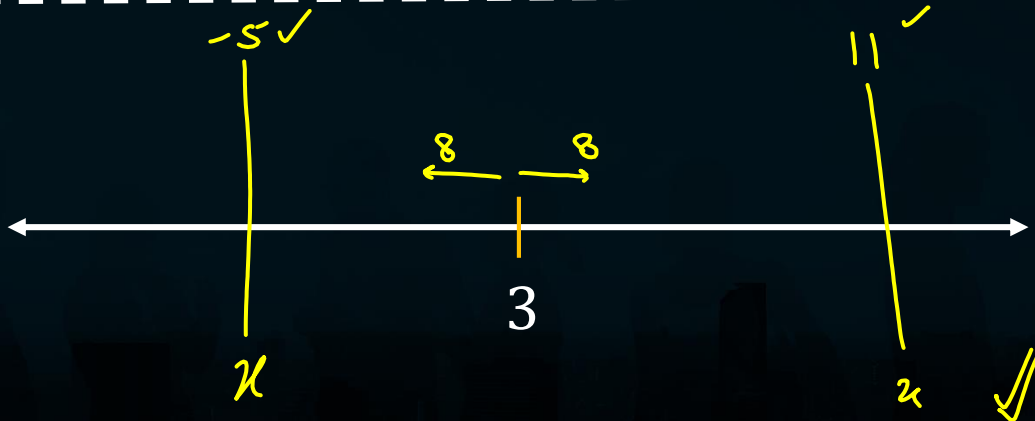
-ve

$$-(x - 3) = 8$$

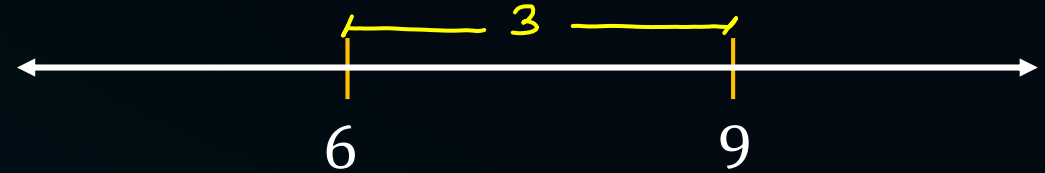
$$x - 3 = -8$$

$$\boxed{x = -5} \checkmark$$

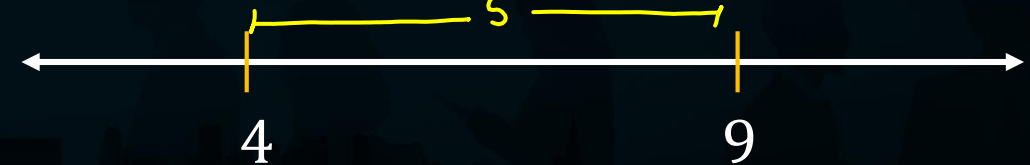
fail



$$|\overset{\checkmark}{9} - \overset{\checkmark}{6}| = |3| = 3$$



$$|4 - 9| = |-5| = -(-5) = 5$$



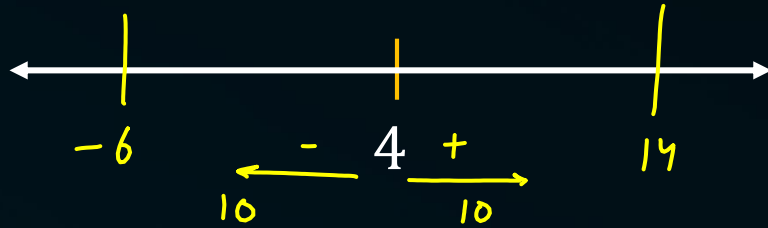


TYPE - 1

Linear Modulus

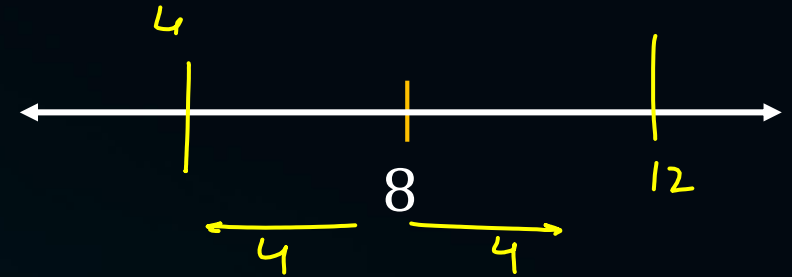
1st Approach $\longrightarrow$ Distance Approach

$$|x - 4| = 10$$



$$\begin{aligned} x &= -6 \\ x &= 14 \end{aligned}$$

$$|\check{x} - \check{8}| = 4$$



$$\boxed{x = 4} \checkmark$$

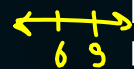
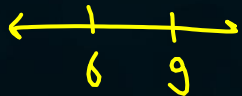
$$\boxed{x = 12} \checkmark$$

$$|5 - 2x| = 7$$

$$|x - 2| = |2 - x|$$

↓
Dist b/n x & 2 ←

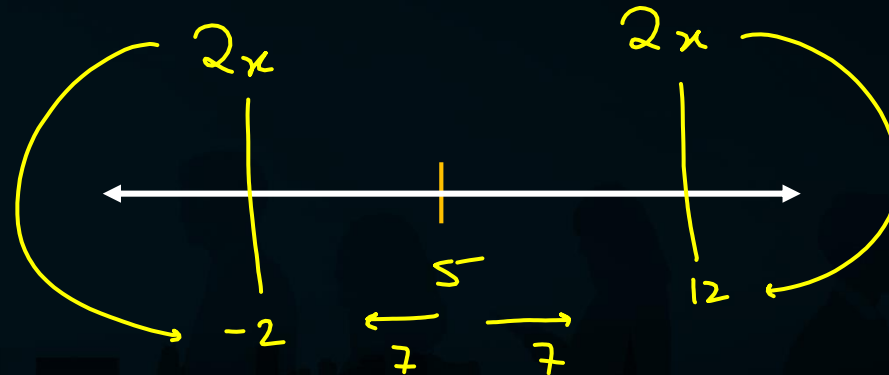
$$|6 - 9| \quad |9 - 6|$$



$$x = -1$$

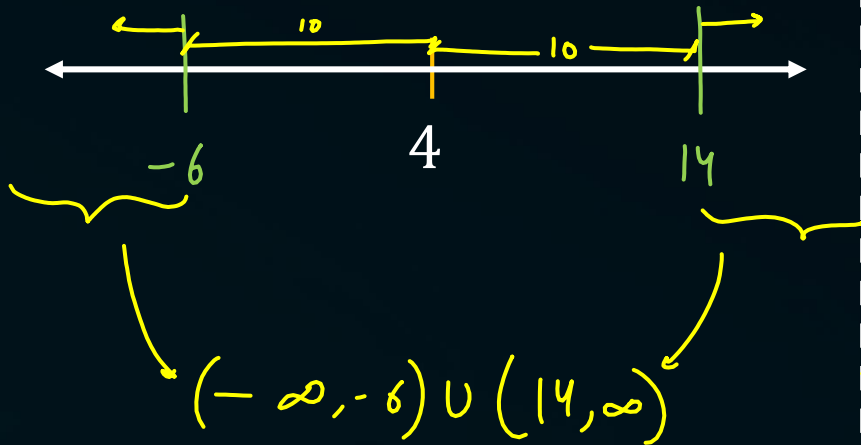
$$|2x - 5| = 7$$

$$x = 6$$



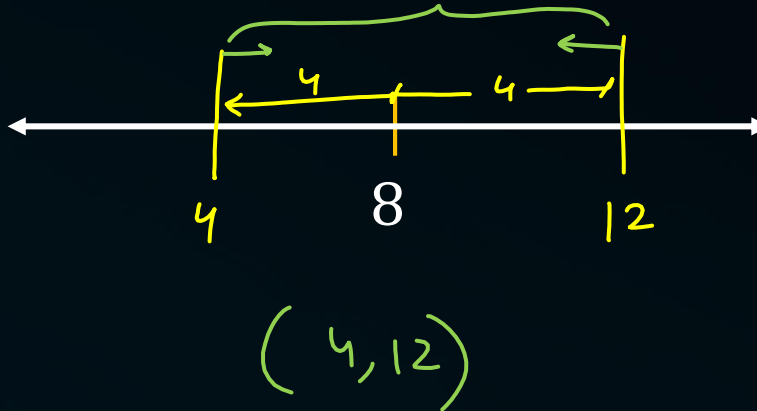
$$|x - 4| > 10$$

$$|x - 4| = 10 \quad \text{HS}$$



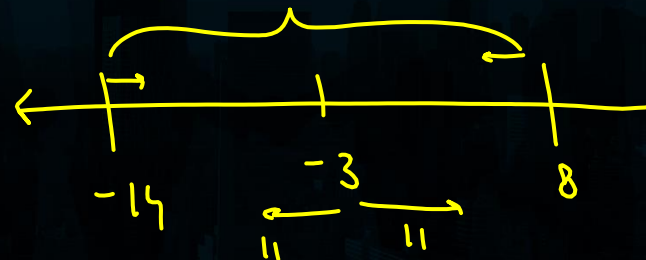
$$|x - 8| < 4$$

$$|x - 8| = 4 \quad \text{HS}$$



$$|x + 3| < 11$$

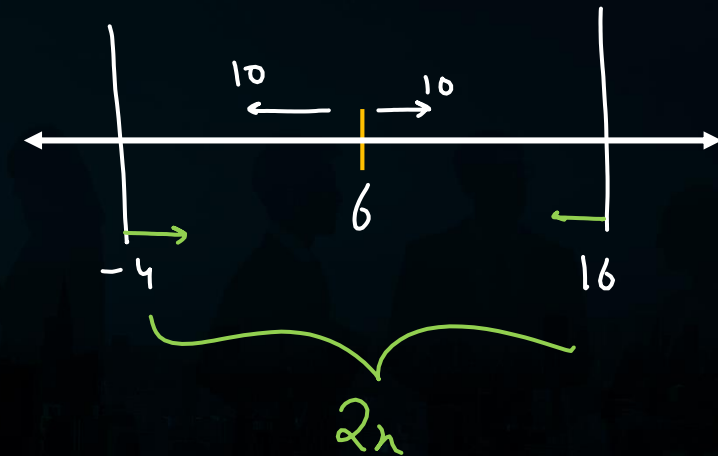
$$|x - (-3)| < 11$$



$$|6 - 2x| \leq 10$$

$$2x \in [-4, 16]$$

$$x \in [-2, 8] \quad \checkmark$$

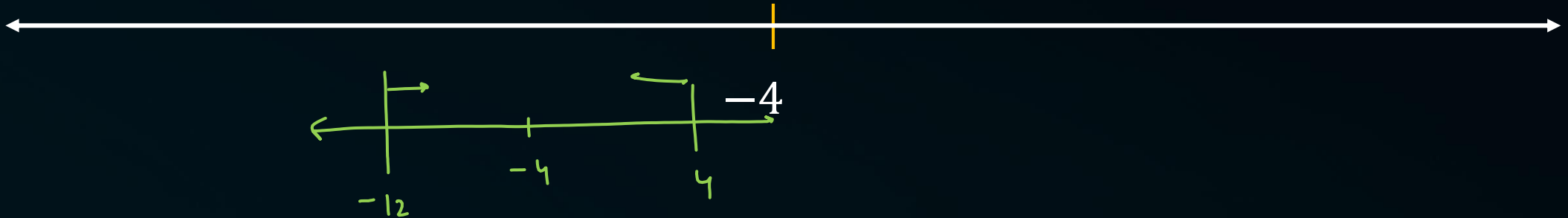


QUESTION - 1



#Q. If $|x + 4| + 2 < 10$, find values of x .

$$|x + 4| < 8$$



$(-12, 4)$

QUESTION - 2



#Q. If $|a - 4| = 5$ and $|2b - 10| = 8$, find the difference between maximum and minimum value of ab .

$$\begin{aligned} 2|b - 5| &= 8 \\ |b - 5| &= 4 \end{aligned}$$

A 72

B 90

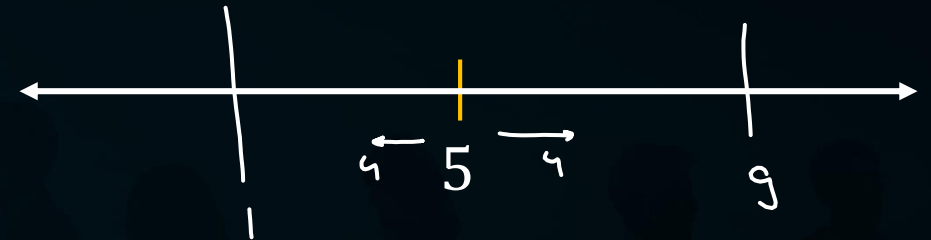
C 81

D none



$$\left. \begin{aligned} a &= -1 \\ a &= 9 \end{aligned} \right\}$$

$$\left. \begin{aligned} ab &= -1 \\ ab &= -9 \\ ab &= 9 \\ ab &= 81 \end{aligned} \right\} 90$$

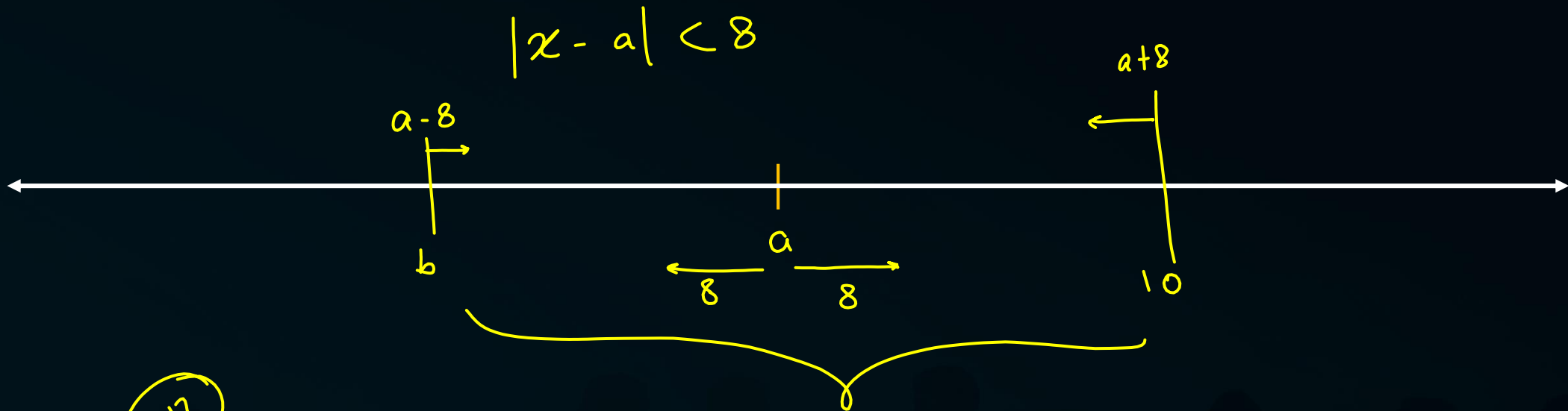


$$\left. \begin{aligned} b &= 1 \\ b &= 9 \end{aligned} \right\}$$

QUESTION - 3



#Q. If $|x - a| - 2 < 6$, has the solution as $(\overset{i.n}{b}, \overset{f.n}{10})$, find the values of a and b .



-12

$$a - 8 = b$$

$$b = 2 - 8 \\ = -6$$

$$a + 8 = 10 \\ \boxed{a = 2}$$

BASIC TYPE 2 : Square of Modulus



- $\check{x}^2 = \check{|x|}^2$

$$x = 2$$

$$2^2 = |2|^2 = 4$$

$$x = -2$$

$$(-2)^2 = |-2|^2 = 4$$

$$|x-2|^2 = (x-2)^2$$

$$|2-x|^2 = (x-2)^2$$

QUESTION - 4



#Q. $(x - 2)^2 = 2|2 - x| + 3$, Find the value of $a - b$ where a is the maximum value of x and b is the minimum value of x .

A

4

B

6



C

2

D

3

$$|2-x|=t$$

$$|x-2|^2 = 2|2-x| + 3$$

$$|2-x|^2 = 2|2-x| + 3$$

$$t^2 = 2t + 3$$

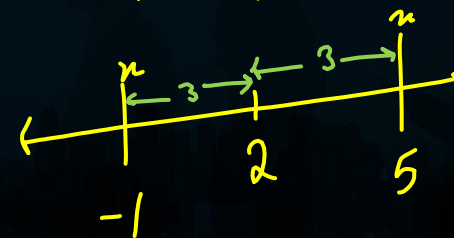
$$t^2 - 2t - 3 = 0$$

$$(t-3)(t+1) = 0$$

$$t = 3$$

$$t = -1 \text{ (crossed out)}$$

$$|2-x| = 3$$



$$x = 5 \\ x = -1 \quad \text{b}$$

$$5 - (-1) \\ = 6$$



TYPE - 2

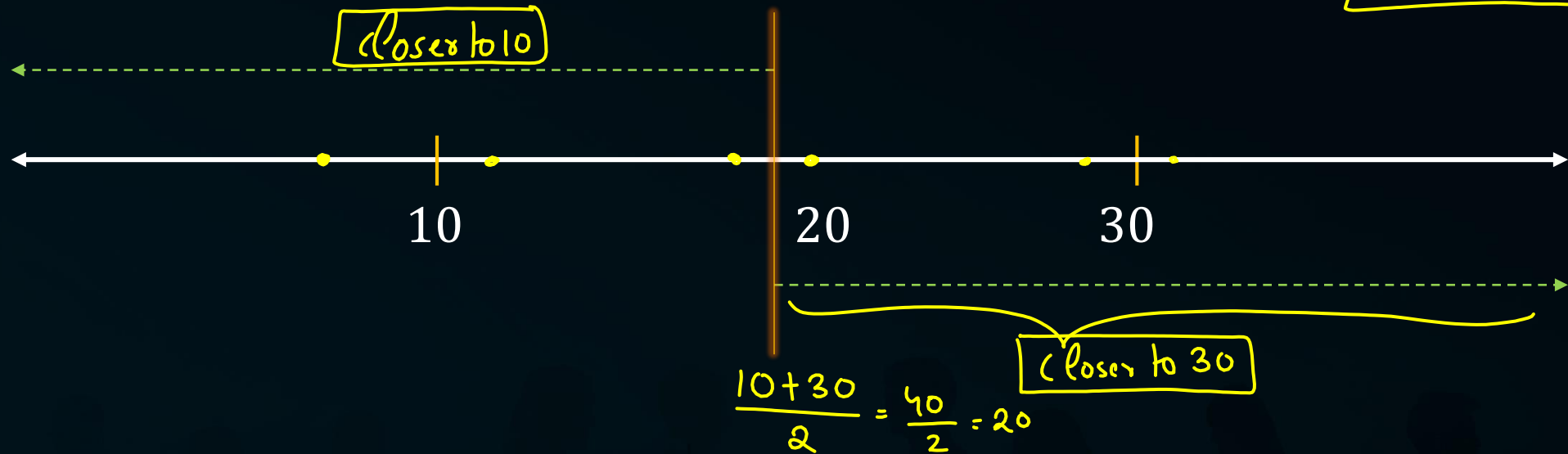
Comparing Modulus

QUESTION - 5



#Q. Find the values of n such that $|n - 10| > |n - 30|$

Dist b/n $n \neq 10 >$ Dist b/n $n \neq 30$
 closer to 30



$(20, \infty)$

QUESTION - 6



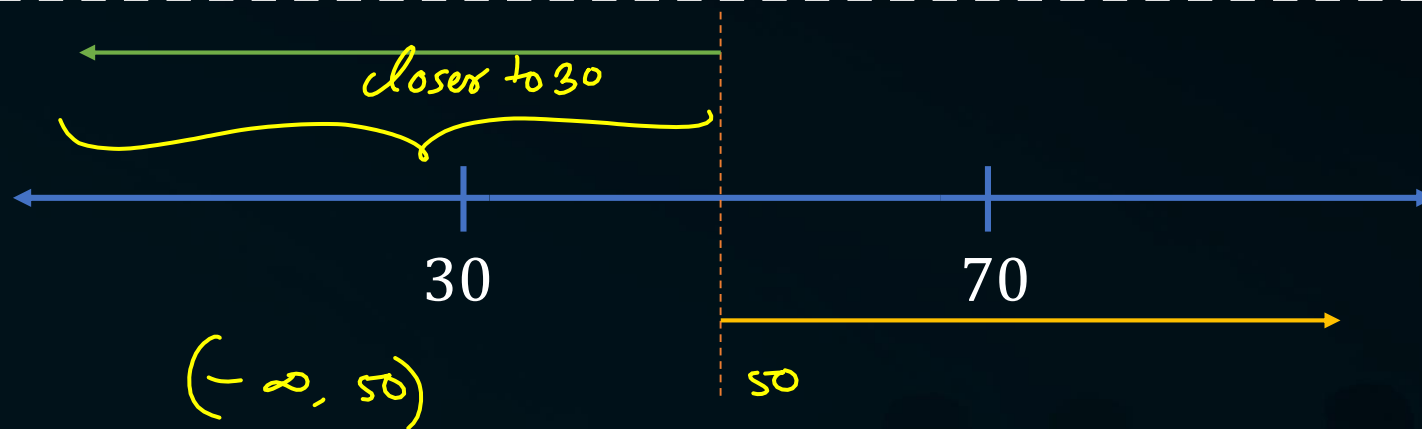
#Q. If n is a multiple of 3, then find the number of possible values of n such that

$$|n - 10| > |n - 70| > |n - 30|$$

$$n \in (40, 50)$$

$$42/45/48$$

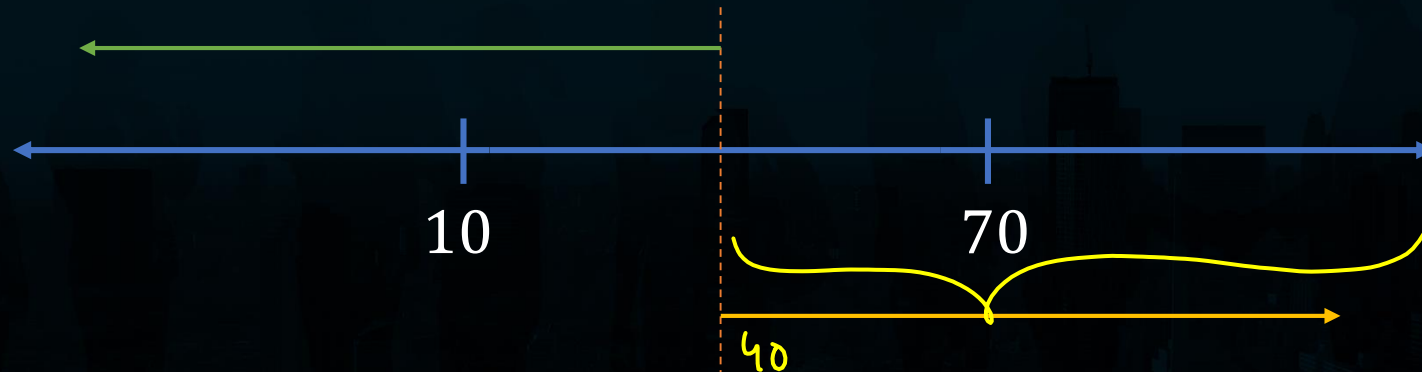
3 ✓



$$|n - 70| > |n - 30|$$

$$n \rightarrow \text{closer to 30}$$

$$(-\infty, 50)$$



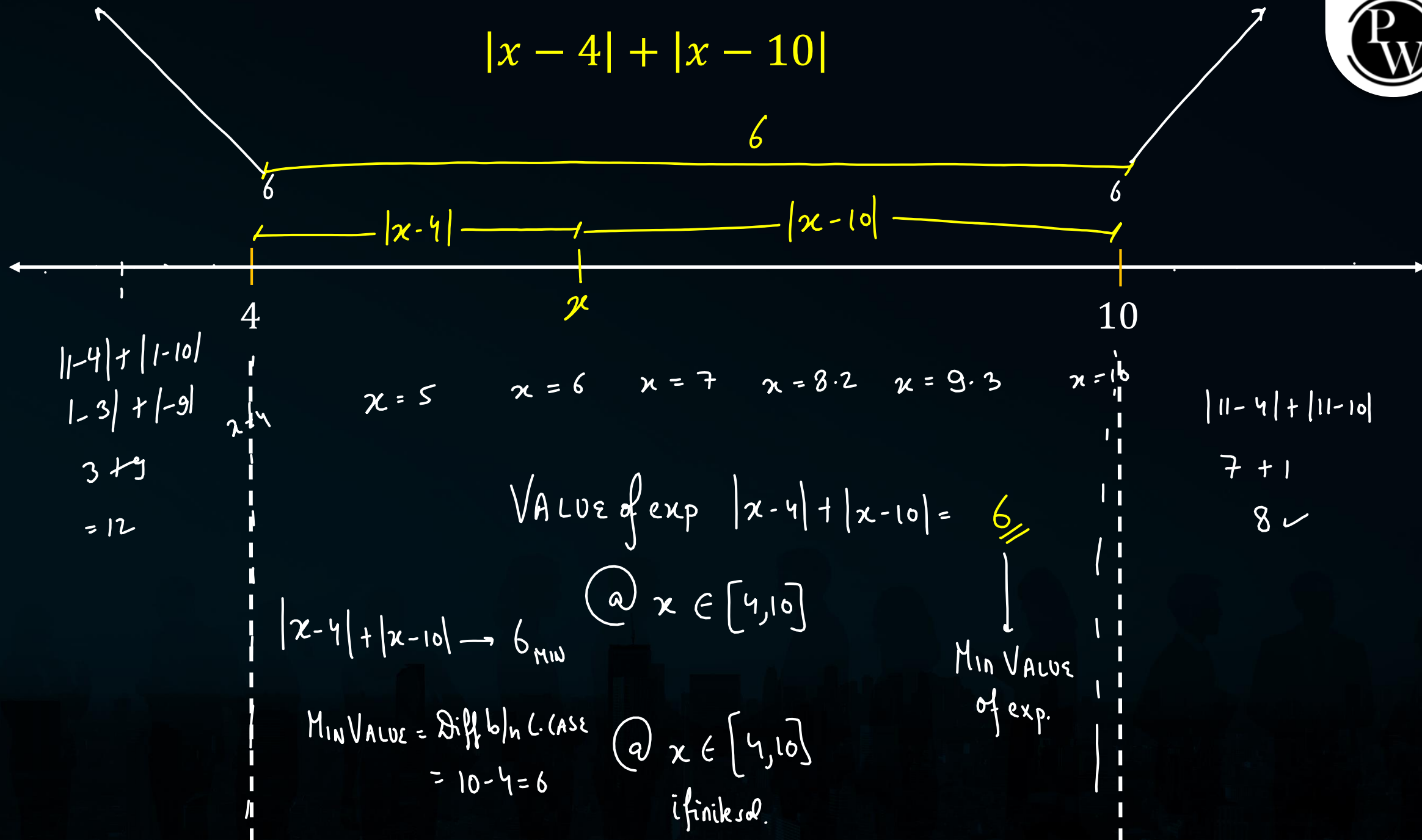
$$|n - 10| > |n - 70|$$

$$n \rightarrow \text{closer to 70}$$

$$(40, \infty)$$



Topics : $|x-a| + |x-b|$ form



$$|x - 2| + |x - 7| = 5$$

$x \rightarrow \text{integral} \rightarrow \text{No. of sol.}$
 $[2, 7]$

6 VALUES

2, 3, 4, 5, 6, 7

* MIN VALUE = 5

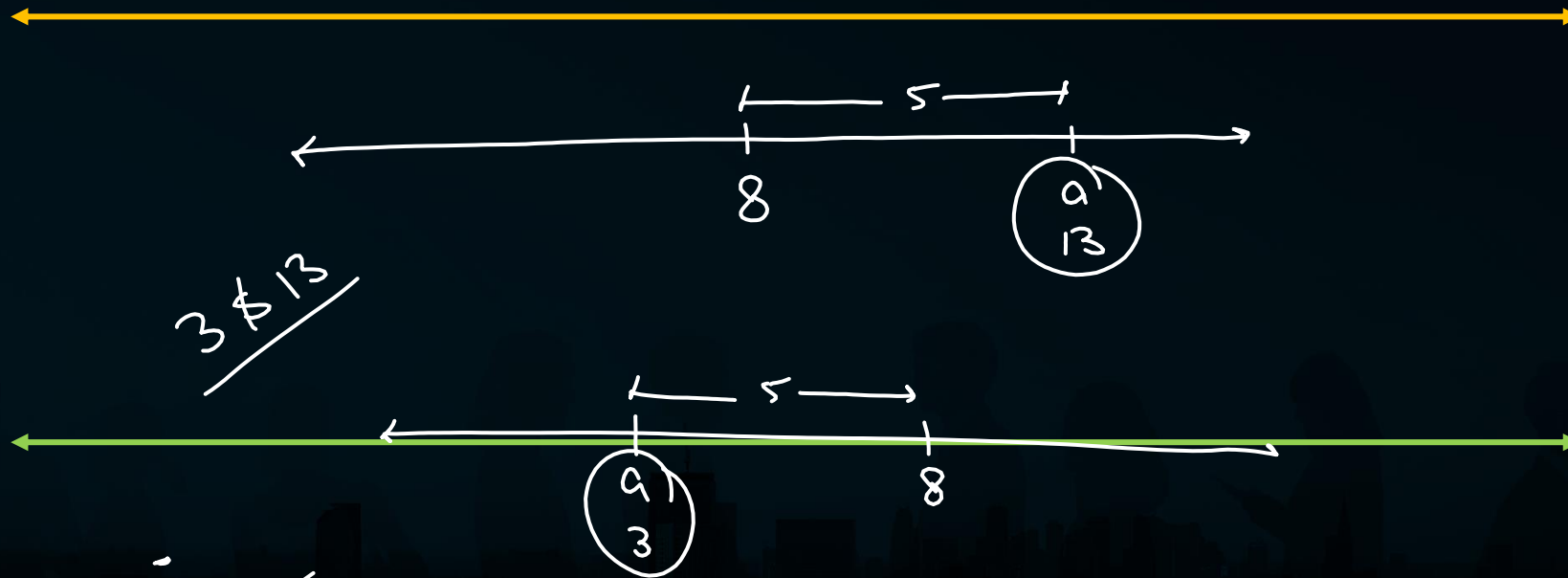
QUESTION - 7



#Q. If the minimum Value of $|x - a| + |x - 8| = 5$. What are the possible values of 'a'.

$\text{Diff} = 5$

$$|a - 8| = 5$$



QUESTION-8 (CAT 2022 - Slot 1)



#Q. The largest real value of ' a ' for which the equation $|x + a| + |x - 1| = 2$ has an infinite number of solution for x is

Diff b/n CR. CASE

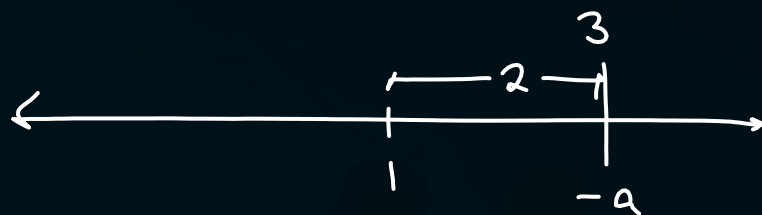
'1' & '-a'

$$-a = 3$$

$$a = -3$$

$$|x - a| \quad \begin{matrix} \nearrow CC \rightarrow a \\ \searrow CC \rightarrow -a \end{matrix}$$

$$|x + a|$$



$$a = 1$$



$$-a = -1$$

$$a = 1 \checkmark$$

$$|x - 4| + |x - 10| = 14$$

$$\frac{8}{2} = 4$$



$$x = 0$$

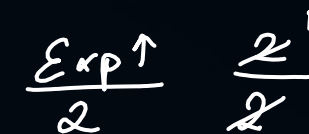
$$\text{Exp} = 6$$

$$(x - 4) + (x - 10) = 14$$

$$2x - 14 = 14$$

$$2x = 28$$

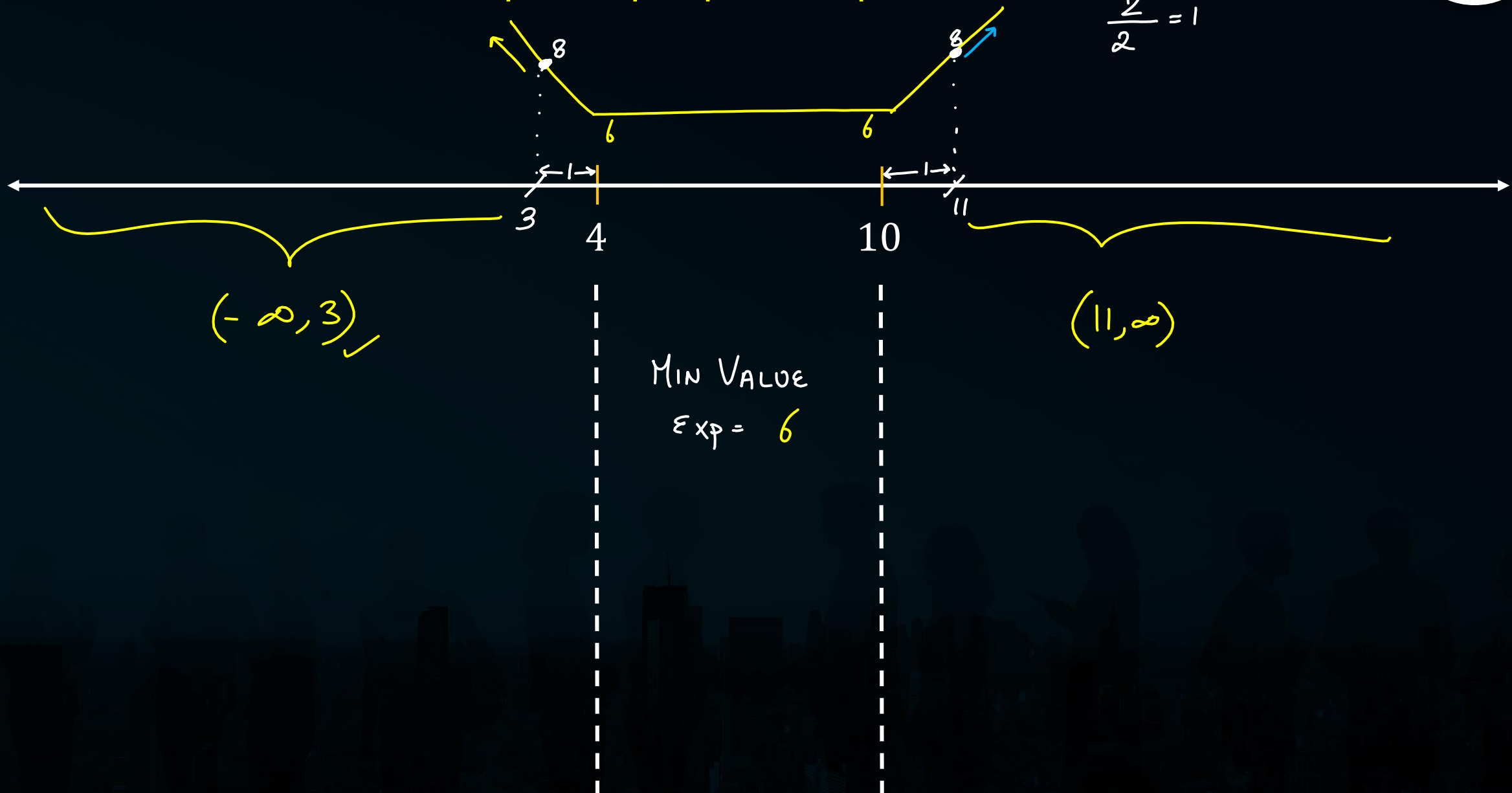
$$x = 14 \checkmark$$



MIN VALUE
 $\epsilon_{xp} = 0$

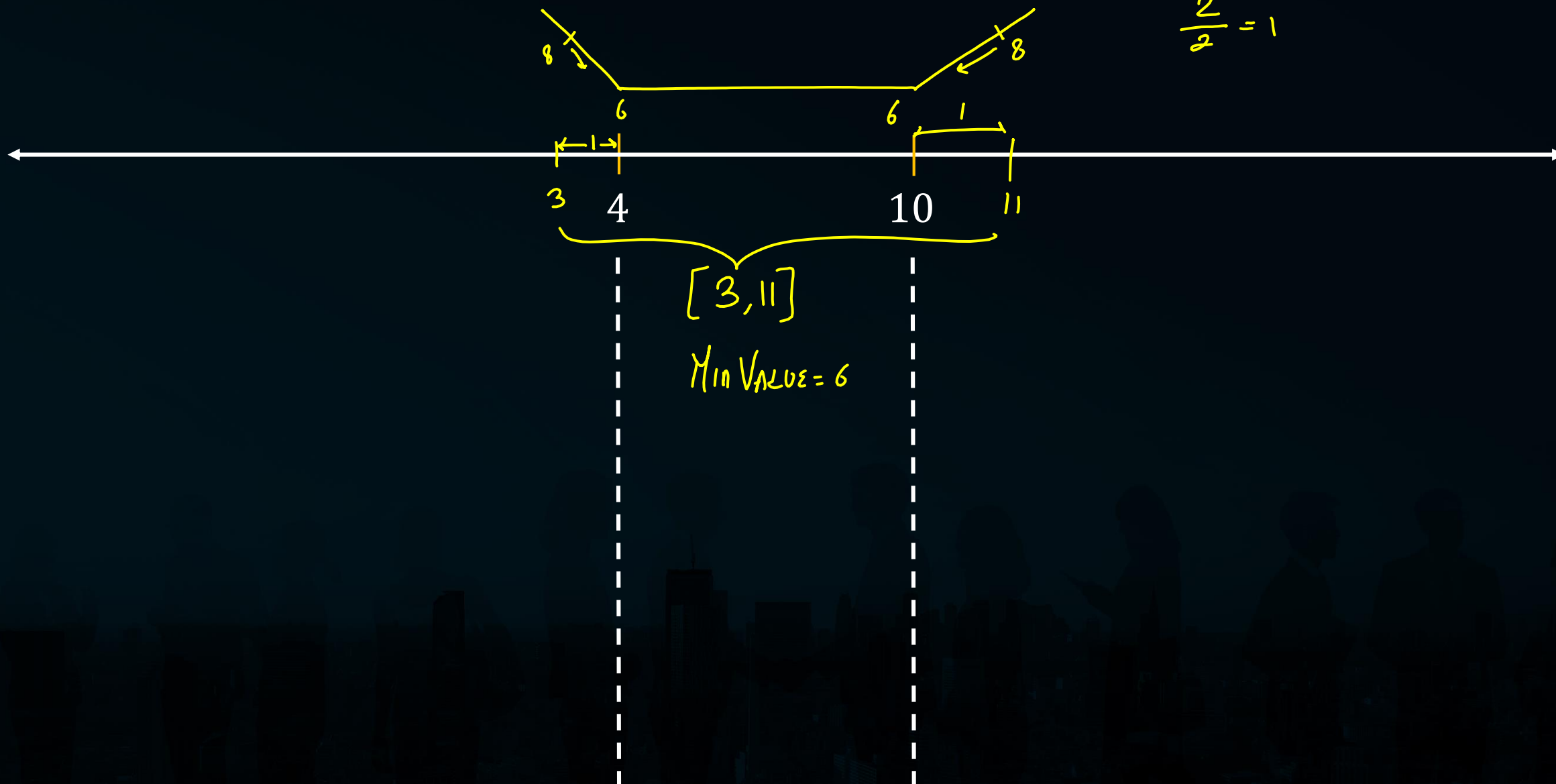
$$|x - 4| + |x - 10| > 8$$

$$\frac{2}{2} = 1$$



$$|x - 4| + |x - 10| \leq 8$$

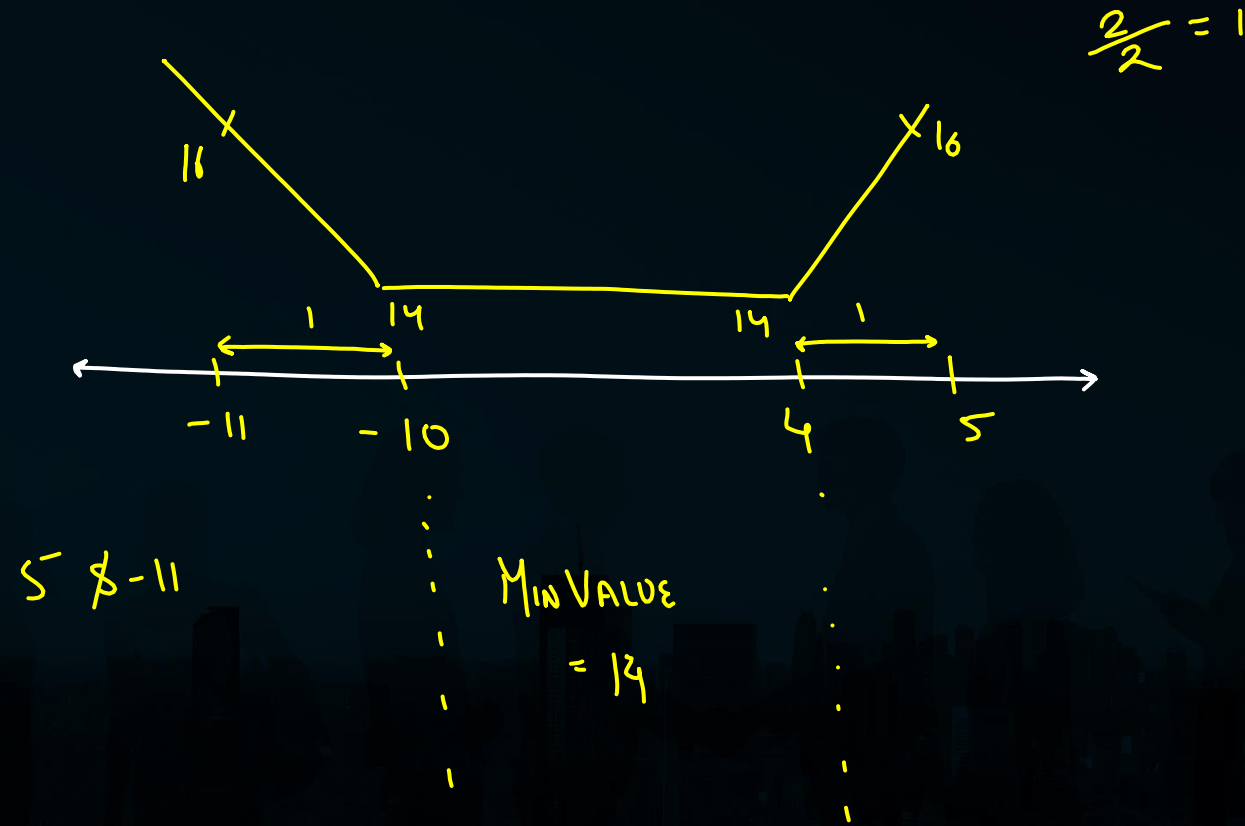
$$\frac{2}{2} = 1$$



QUESTION-9



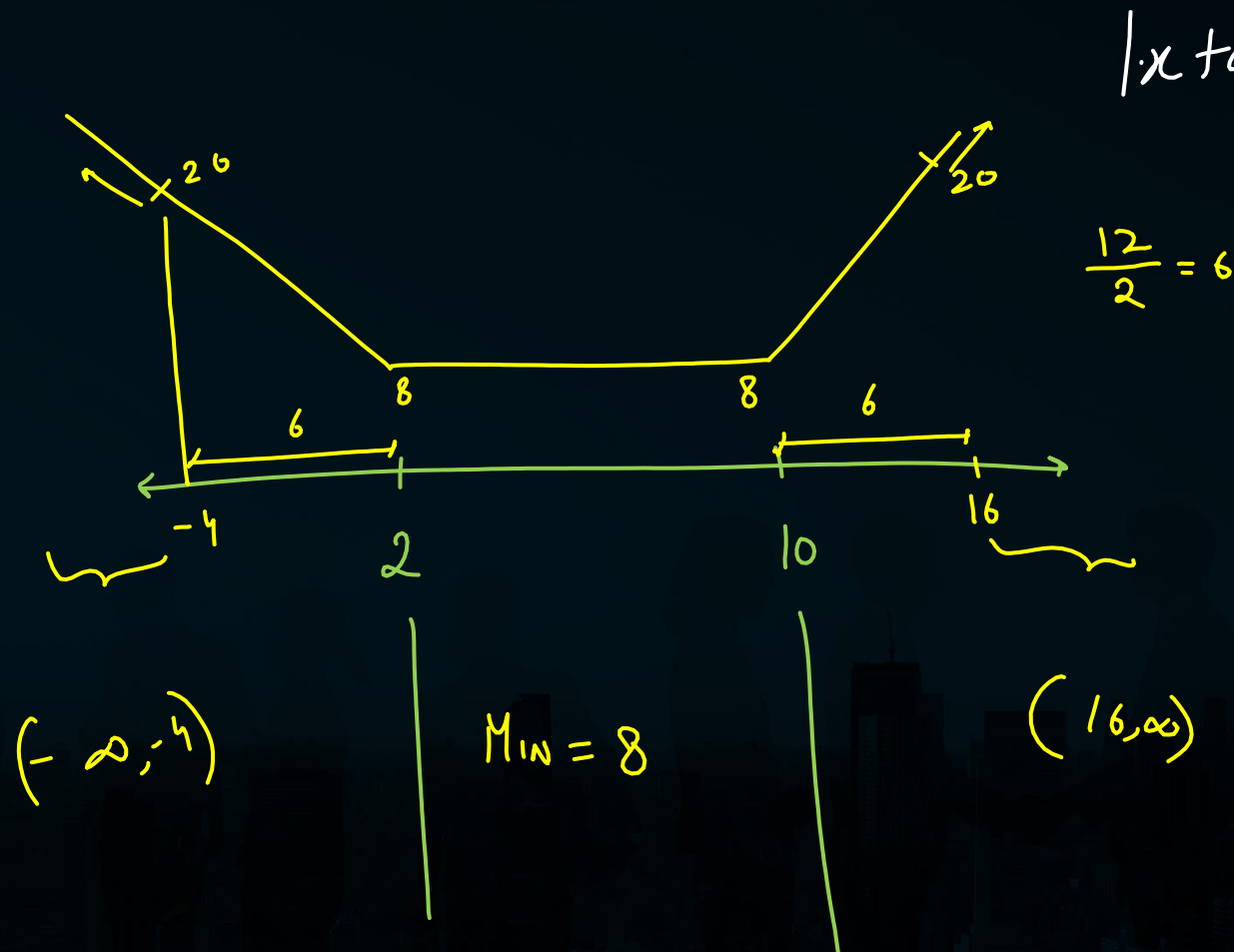
#Q. If $|x - 4| + |x + 10| = 16$, find values of x .



QUESTION-10



#Q. If $|x - 2| + |x - 10| > 20$, find values of x .



$$|x+a| + |x+b| \geq$$

$$|x-a| >$$

$$|x-a| + b >$$

$$|x-a| > |x-b|$$



Type - 3

Multiple Modulus

QUESTION - 11



#Q. Find the minimum Value of

$$|x - 10| + |x - 30| + |x - 50| + |x - 80| + |x - 90|.$$

$|90 - x|$ with an arrow pointing to the last term.

$$40 + 20 + 0 + 30 + 40 \\ = 130 \checkmark$$



$$(x = 50) \\ \swarrow \text{Min}$$

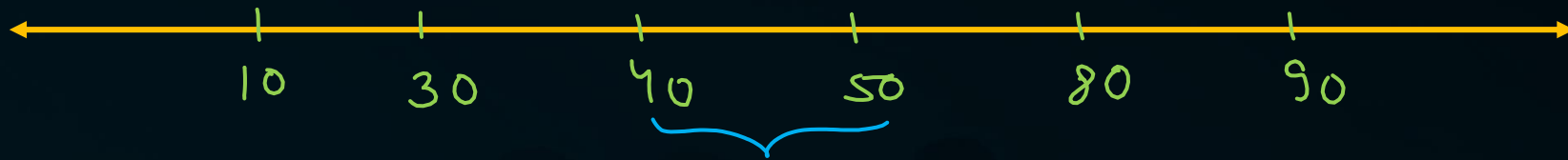
QUESTION - 12



#Q. Find the minimum Value of
 $|x - 10| + |x - 30| + |x - 40| + |x - 50| + |x - 80| + |x - 90|$.

30 10 0 10 40 50

140



at $x \in [40, 50]$

Min Value of exp = 140



Topics : Modulus – Miscellaneous

QUESTION - 13



#Q. $(|x| - 4)(|y| + 3) = 13$, Find the number of ordered pairs satisfying this equation given that x and y are integers.

$$(|x| - 4) (|y| + 3) = 13$$

Min value $\rightarrow 3$

Approach

✓ 4 ordered

$$\begin{array}{r} 1 \qquad 13 \\ \hline 13 \qquad 1 \\ \hline 1 \qquad -13 \\ \hline -13 \qquad -1 \end{array}$$

$$\begin{aligned} |y| + 3 &= 1 \\ |y| &= -2 \end{aligned}$$

$$|x| - 4 = 1$$

$$|x| = 5$$

$$x = \pm 5$$

$$\begin{aligned} (5, 10) & \quad (-5, 10) \\ (5, -10) & \quad (-5, -10) \end{aligned}$$

$$|y| + 3 = 13$$

$$|y| = 10$$

$$y = \pm 10$$

Ques 15 : THE OG TRADITIONAL METHOD



#Q. $x^2 + 3x = |2x - 14|$, Solve for x

Number line diagram showing critical points -7 and 2 . The region $x < -7$ is marked with a double checkmark. The region $-7 < x < 2$ is marked with a checkmark. The region $x > 2$ is marked with a double checkmark.

Case 1: $x < -7$

$$x^2 + 3x = -(2x - 14)$$
$$x^2 + 3x + 2x - 14 = 0$$
$$x^2 + 5x - 14 = 0$$
$$(x + 7)(x - 2) = 0$$
$$\left\{ \begin{array}{l} x = 2 \\ x = -7 \end{array} \right\} \checkmark$$

Case 2: $x > 2$

$$x^2 + 3x = 2x - 14$$
$$x^2 + x + 14 = 0$$
$$D < 0$$

No soln

MOD KA GOD



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Functions Complete Modulus in One Shot | CAT 2024 Quant Preparation

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10 chapters Introduction | What is Modulus | Type 1 Linear Modulus | Type 2 Comparing Modulus | T

MUST WATCH 2:04:34 – 2:31:22



THANK
You

